

Embedded Systems Final Project

ECE 5780

Alan Banner, Alex Gray, Jessika Jimenez

Bill of Materials:

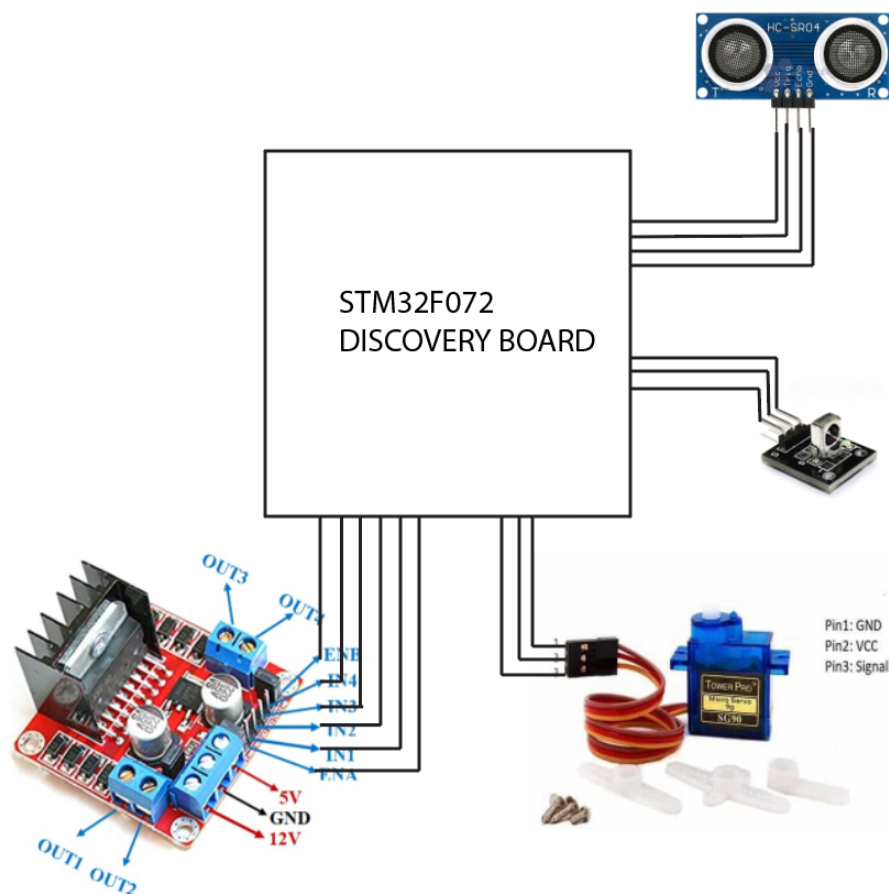
- Rover chassis - cost: (already own)
- Ultrasonic sensor HC-SR04 - cost: (already own + 2 spare)
- Servo SG90 - cost: (already own)
- Infrared sensor TL1838 - cost: (already own) (ended up not using)
- DC motors x4 - cost: (already own)
- Discovery board - cost: (already own)
- Bluetooth module - cost: (\$10.39)
- Accelerometer - cost: (\$10 for 3)
- Power module - cost: (already own)

Issues:

- Power -> Rover battery packs
- Sensing -> Ultrasonic sensor (ended up not using infrared sensor)
- Control -> Controlled over USART with GUI
- Motion -> Motors and wheels

System Design:

- Diagram of the rover's physical component connections:



Project Schedule Planning:

1. Week 0 - 2/21:
 - a. Complete Mini Project: Proposal assignment (Due Fri 3/3)
2. Week 1 - 3/2:
 - a. Await project feedback.
 - Acquire all project components [DONE]
 - Start rover GUI [DONE]
3. Week 2 - 3/6:
 - a. Begin work for Milestone 1
 - i. Next goal: Ensure the rover motor works/the rover can move freely.
 - Understand pinout/wire for rover [DONE]
 - Connect up the STM32 board to the rover [DONE]
 - Begin rover GUI [DONE]
4. Week 3 - 3/16:
 - a. Complete Milestone 1 (Due Fri 3/17)
 - i. Next goal: Begin working on the ultrasonic sensor & connect accelerometer on the rover.
 - Be able to move rover drivers [DONE]
 - Get GUI to control rover via UART [DONE]
5. Week 4 - 3/23:
 - a. Complete Milestone 2 (Due Fri 3/24)
 - i. Next goal: Have fully working ultrasonic sensor for rover.
 - Wire up accelerometer [DONE]
 - Begin wiring/coding ultrasonic sensor [DONE]
6. Week 5 - 3/28:
 - a. Complete Milestone 3 (Due Fri 3/31)
 - i. Next goal: Finish ultrasonic and accelerometer things.
 - For ultrasonic sensor, get timer interrupt & get return value [DONE]
 - Setup state machine for ultrasonic sensor [DONE]
 - Order bluetooth adapter for serial communication to rover [DONE]
7. Week 6 - 4/6:
 - a. Complete Milestone 4 (Due Fri 4/7)
 - i. Next goal: Have GUI update on accelerometer & ultrasonic values.
 - Finish ultrasonic sensor things (scaling) [DONE] (good enough)
 - Connect accelerometer to rover [DONE]
 - Connect bluetooth adapter to rover [DONE]

8. Week 7 - 4/13:
 - a. Complete Milestone 5 (Due Fri 4/14)
 - i. Next goal: Finish coding and testing.
 - Connect servo motor for ultrasonic sensor [DONE]
 - Use systick to update servo movement [DONE]
 - Change accelerometer to do interrupt instead of who_am_i [DONE]
9. Week 8 - 4/18 & 4/20:
 - a. Complete Milestone 6 (Due Fri 4/21)
 - i. Next goal: Do well at the rally demo.
 - Have GUI image update on ultrasonic values [DONE]
 - Have GUI image update on accelerometer values [DONE]
 - Buy replacement bluetooth serial adapter [DONE]
 - Ensure working accelerometer interrupt [DONE]
 - Complete rover testing [DONE]
10. Week 9 - 5/3:
 - a. Prep for Rally Demo (Due Wed 5/3)

Risks/Unknowns/Potential Problems and Solutions:

- Risks/Potential Problems:
 - Old Battery may not hold a charge -> have replacement batteries on hand
 - Part failures -> have spare parts on hand
 - Sensor malfunction -> use redundant sensor types (infrared and ultrasonic sensors)
- Unknowns:
 - How to use/program the infrared and ultrasonic sensors -> review component datasheets
 - What the I/O pins are on the discovery board -> determine how many and what pins are required, based on component and discovery board datasheets
 - How to control the rover -> determine control inputs and feedback required for GUI design

Additional Notes For Us:

1. Pins PB7, PB6, PB5, PB4, PB3, PD2 are for motor driver
 - a. (ENA, IN1, IN2, IN3, IN4, ENB)
2. Pin PA8 is for servo for ultrasonic sensor
 - a. (PWM)
3. Pins PA15 (5V tolerant, TIM2 interrupt), PB8 are for ultrasonic sensor (TIM16)
 - a. (Echo Pulse Output [green], Trigger Pulse Input[yellow])
4. Pins PC10, PC11 are for USART3
 - a. (TX, RX)
5. Pins PB10, PB11, PA0 are for accelerometer
 - a. (SCL, SDA, interrupt)

In1 (PB6)	In2 (PB5)	In3 (PB4)	In4 (PB3)	direction
0	0	0	0	stop
0	1	1	0	forward
1	0	0	1	backwards
0	1	0	1	left
1	0	1	0	right

(^ To move the rover wheels. State machine for the ultrasonic sensor v)

